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Data Article

Data on physical impacts and hydrogeochemical assessment of inactive/abandoned mines in and around Southwestern parts of the Cuddapah basin using a conceptual site model (CSM)

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A B S T R A C T

The conceptual site model (CSM) has been designed for Inactive/abandoned mines in the SW part of Cuddapah basin, which gives an overview of Inactive/abandoned mine characterization in terms of physical impacts such as vertical openings, dangerous impoundments, location of steep slopes, waste processing facilities, steep portal abandoned barite mine and assessment of groundwater analytical data. To evaluate the groundwater quality and its suitability for domestic, drinking purposes, 44 groundwater samples were collected from the southwestern part of the Cuddapah basin, were examined for major cations and anions. The suitability of groundwater for drinking purposes is assessed by comparing with the World Health Organization (W.H.O) and Indian standards (IS).

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Specifications Table

Subject area	Chemistry
More specific subject area	Environmental Geochemistry
Type of Data	Tables and figures
How data was acquired	The conceptual site model (CSM) was designed for the identification of vertical openings, dangerous impoundments, location of steep slopes, waste processing facilities, and analysis of groundwater. 44 Groundwater samples were collected from bore wells in the southwestern part of Cuddapah basin, Y·S.R District, A.P; pH and EC, TDS (Conductivity cell CD-10) are determined with help of water analyzer 371 field kit respectively; Total Hardness, Ca^{2+}, Mg^{2+}, $CO_3{}^{2-}$, $HCO_3{}^{-}$ and Cl^{-} were determined using titrimetry as laboratory followed standard methods (APHA 2012); F^{-} is determined using ion-selective electrode (Orion 4 star ion meter, Model: pH/ISE).
Data format	Raw, analyzed
Parameters for data collection	All the parameters were analyzed according to standard procedures for the examination of groundwater [1] (Table 1)
Description for data collection	Major cations, anions levels in drinking water were determined and compared with WHO and BIS drinking water standards [2]
Data Source Location	Southwestern part of Cuddapah basin, A.P India (Fig. 1)
Data accessibility	Data is available in the article.

Value of the Data

- The conceptual site model (CSM) helps to locate vertical openings, dangerous impoundments, steep slopes, waste processing facilities of Inactive/abandoned mines in the study area.
- Quality analysis data in this study area can help to understand the quality of groundwater and calls for the need for constant monitoring of remediation technologies.
- Due to limited studies in the study area, the data of this study can help to better understand the environmental and human health problems in the inactive/abandoned mine areas and provide a scope for further studies.

1. Data description

1.1. Study area

The study area falls under the southwestern (SW) part of the Cuddapah basin shown in Fig. 1 and covering four mandals; Lingala, Pulivendula, Vempalli, and Vemula. Lingala, Pulivendula, Vempalli, and Vemula. It lies between latitude 14° 18′ 0″ N to 14° 28′ 0″; longitude 78° 0′ 00″ E 78° 30′ 0″ falls in Topo sheet no 57 J/02, 57J/03, 57J/04,57J07 (Fig. 1). The study area consists of purple shale, massive limestone, intraformational conglomerate, dolostone (uraniferous), shale, quartzite, cherty limestone and basic intrusive in Papaghni and Chitravati groups belongs to Lower Cuddapah supergroup [3]. The major geomorphic units of the study area are Denudational hills, Pediment & Pediplain. The soil types of the study area are black, alluvial, brown and mixed soils. The average annual rainfall is 600–650 mm and the average temperature varies from 20.4 °C in December to 43.2 °C in April [4].

1.2. Analytical data

The conceptual site model (CSM) designed for simply approaching for inactive/abandoned mines investigation as shown in Fig. 2. Analytical data of the groundwater samples given in Table 1. The data of physicochemical parameters of individual groundwater samples and statistical parameters like mean, median and mode data are given in Table 2. Graphical representation of statistics of

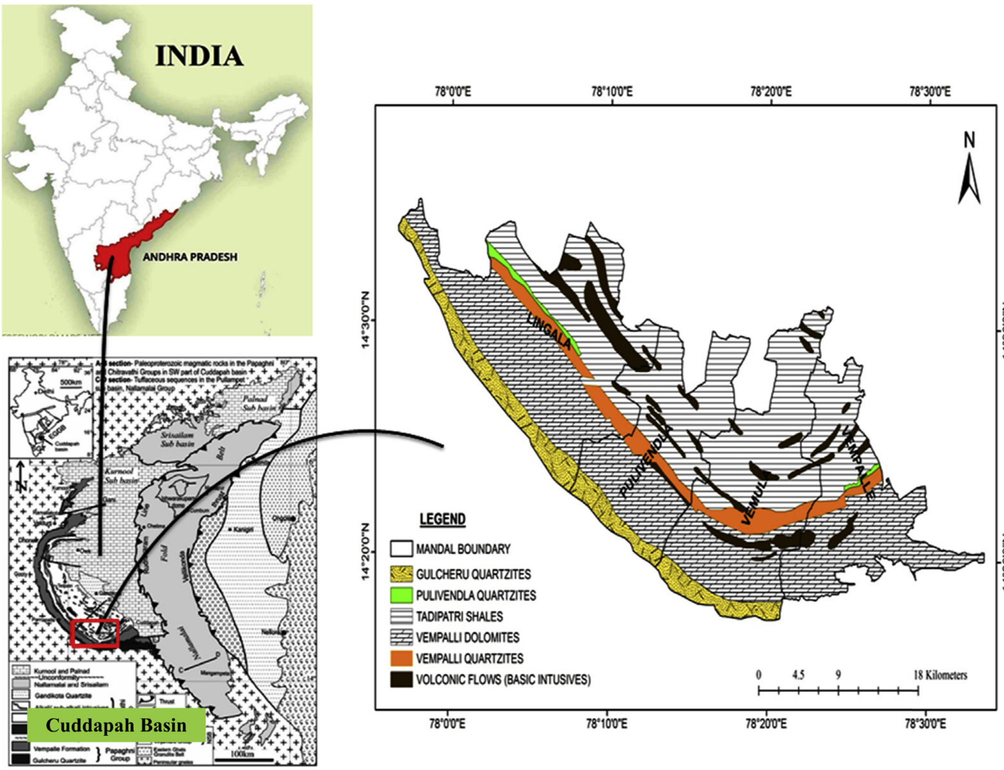

Fig. 1. Location map of the study area (by Sesha Sai et al. present work; modified after Nagaraja Rao et al., 1987).

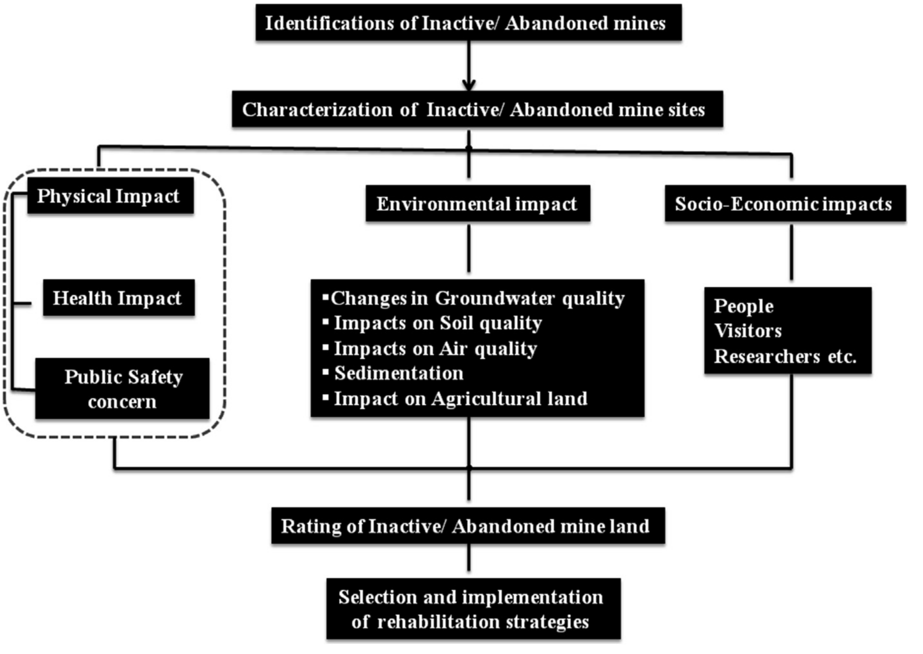

Fig. 2. CSM model for simplified approaching for dealing with inactive/abandoned mines.

physicochemical parameters is shown in Fig. 3. Fluoride classification of groundwater in the study area 6.8% of people fall under the high-risk category in the view of dental and skeletal fluorosis as shown in Table 4. The physical and chemical impacts due to inactive/abandoned mine site prospects data of the study area were given in Table 5. The frequency distribution of fluoride concentration for risk evaluation in groundwater is shown in Fig. 4. Correlation among physicochemical parameters of the Groundwater samples as depicted in Table 3. As from Fig. 5 shows a strong positive correlation is observed from the correlation coefficient values between EC and TDS (0.9949), EC- TH (0.6685), TH - TDS (0.65), TH – Cl- (0.57) are positively correlated. Dangerous slope and high walls of abandoned clay mine present near Lingala village attractive nuisance and is located within close distance to a populated area, the public road is depicted in Fig. 6 [5–15].

2. Experimental design, materials, and methods

The conceptual site model (CSM) was developed for a simplified approach to the assessment of the physical and environmental impacts of the Inactive/abandoned mines in the SW part of the Cuddapah basin. The fieldwork began with the identification and location of all the Inactive/abandoned mines in the SW part of the Cuddapah basin. 44 Groundwater samples were collected in and around inactive/ Abandoned mines in the Southwestern part of the Cuddapah basin, Andhra Pradesh during September 2018 and taken necessary precautions to avoid contamination. All the groundwater samples were collected in two-liter pre-cleaned and well-dried polyethylene bottles and analyzed electrical conductivity (EC), pH, total dissolved solids (TDS), major cations and anions, adopting the standard methods APHA 2012 [16]; pH and EC, TDS (Conductivity cell CD-10) are determined with help of water

Table 1
Analytical data for the groundwater samples from the study area.

S·NO	Village	pH	EC μS/cm	TDS mg/L	TH mg/L	Ca^{2+} mg/L	Mg^{2+} mg/L	CO_3^{2-} mg/L	HCO_3^- mg/L	Cl^- mg/L	F^- mg/L
1	VS1	8.4	823	382	160	24	34	12	293	28	0.4
2	VS2	8.8	390	203	160	8	10	24	73	36	0.24
3	VLP3	8.1	1450	750	300	48	59	24	146	312	0.27
4	GLD 4	8.5	2590	1340	280	16	16	24	317	454	1.27
5	GLD 5	8.2	369	188	100	40	83	12	122	43	0.62
6	GLD 6	8.8	1500	800	200	16	20	36	317	36	2.06
7	VEM 7	8	2150	1100	280	16	44	24	293	78	0.73
8	GON 8	8.4	1040	540	200	16	15	12	220	142	0.66
9	VEM 9	8.1	634	327	180	16	40	24	195	135	0.46
10	GON 10	7	1070	555	220	40	58	12	268	21	0.79
11	VKP 11	8.7	1430	740	340	24	39	36	244	220	0.71
12	GUN 12	8.2	1760	910	140	24	20	24	219	135	1.6
13	GON 13	8.6	2170	1120	520	56	40	36	244	369	1.22
14	RAN 14	8.9	1450	750	180	16	78	48	366	85	2.55
15	PRN 15	8.6	1180	610	140	56	40	24	317	57	3
16	CGU 16	7.8	1710	880	280	16	63	60	24	234	0.87
17	CGU 17	8.5	826	423	200	40	49	24	122	71	1.6
18	KGB 18	8.8	1220	630	160	40	68	36	268	120	0.88
19	BMP 19	8.1	1000	516	260	88	39	24	171	106	0.8
20	MDP 20	7.9	1150	590	360	40	82	12	195	85	0.66
21	VEM 21	8.1	811	421	200	40	68	24	195	50	0.87
22	BSP 22	8.1	1070	550	240	24	34	24	48	99	0.92
23	BSP 23	7.7	967	500	260	24	63	36	38	78	0.74
24	VLP 24	8.3	792	409	200	40	63	12	39	43	0.66
25	VLP 25	8.4	774	400	200	40	117	36	40	36	1.19
26	DUGP 26	8.2	1170	600	300	32	49	12	41	85	1.55
27	ALVP 27	8.4	247	127	80	24	44	14	98	35	0.38
28	ALVP 28	7.6	3960	2050	400	56	45	12	293	618	0.56
29	ALVP 29	8.4	233	121	120	24	117	0	73	28	0.55
30	VEM 30	8.4	639	329	440	40	10	12	122	57	0.75
31	VEM 31	8.1	2100	1090	340	40	24	12	244	291	1.08
32	VEMO 32	8.6	977	502	260	24	44	24	268	64	0.81
33	KUPP 33	8.2	1520	790	320	48	24	12	244	149	0.52
34	KUPP 34	7.8	1980	1020	360	24	15	24	195	277	0.56
35	THP 35	8.7	1850	960	300	32	39	24	342	20	0.87
36	CHRP 36	7.9	1050	540	240	40	54	24	221	92	0.91
37	GIDVP 37	7.6	1320	680	380	56	63	24	122	49	0.83
38	BKP 38	7.7	961	494	280	48	23	24	195	57	0.67
39	BKP 39	7.8	2030	1050	400	64	77	24	196	206	0.51
40	AMP 40	7.9	2260	1017	460	32	87	48	244	255	1.37
41	AMBP 41	6.1	390	250	120	64	29	40	230	28	0.6
42	NGP-42	7.9	410	262	80	104	53	30	220	64	0.7
43	NGP 43	7.8	320	205	120	88	38	26	200	78	0.2
44	MKGP-44	8	360	230	100	80	58	30	220	28	0.7

Table 2
Statistical parameters of Southwestern part of Cuddapah basin.

	Min	Max	Mean	St.Dev	CV	Median
pH	6.1	8.9	8	0.5	6	8
EC μS/cm	233	3960	1229	741	60	1070
TDS mg/L	121	2050	635	375	59	552
TH mg/L	80	520	246	107	43	240
Ca^{2+} mg/L	8	104	39	21	55	40
Mg^{2+} mg/L	10	117	48	25	52	44
CO_3^{2-} mg/L	0	60	24	11	47	24
HCO_3^- mg/L	24	366	194	92	47	209
Cl^- mg/L	20	618	126	127	100	78
F^- mg/L	0.2	3	0.9	0.5	62	0.7

Table 3
Correlation of water quality parameters.

	pH	EC	TDS	TH	Ca^{2+}	Mg^{2+}	CO_3^{2-}	HCO_3^-	Cl^-	F^-
pH	1									
EC	0.006597	1								
TDS	−0.00364	0.997459*	1							
TH	−0.03904	0.668526*	0.650018*	1						
Ca^{2+}	−0.3805	−0.1467	−0.12207	−0.0536	1					
Mg^{2+}	−0.03399	−0.14743	−0.16732	−0.08673	0.103267	1				
CO_3^{2-}	−0.03206	0.135555	0.130064	0.069014	−0.0148	0.119485	1			
HCO_3^-	0.110248	0.432164	0.440017	0.077633	0.056673	−0.27855	0.095566	1		
Cl^-	−0.02906	0.821947*	0.82518*	0.570965	−0.01735	−0.16565	0.089897	0.224608	1	
F^-	0.343733	0.207242	0.202335	−0.02051	−0.13886	0.037513	0.320365	0.321828	−0.03485	1

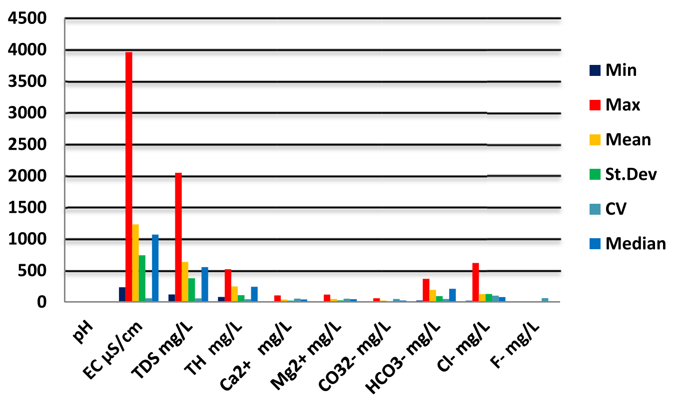

Fig. 3. Statistics of physico-chemical parameters.

Table 4
Fluoride classification of groundwater of the study area.

F- mg/L	Health Impact on humans	Frequency
<0.5	Dental caries	15.90%
0.6−1.5	Required levels for human	70.45%
1.6−2	Dental fluorosis	68.80%
2.1−3	Dental and skeletal fluorosis	6.80%
>3	leads to skeletal fluorosis	–

Table 5
Distance from rural areas to inactive/abandoned mine sites.

Inactive/abandoned mine	Village	Latitude	Longitude	Distance from villages	Distance from agriculture land	Impacts	
						Physical	Chemical
Barite	V. Kottapalli	14.346	78.351	50 m	10 m	a. Steep slope high wall b. Dangerous impoundment c. Loose surface d. Vertical openings (no barricaded)	Effect on groundwater quality and soil quality
	Vemula	14.355	78.310	15 m	10 m		
	Vemula	14.367	78.345	20 m	30 m		
	Vempalli	14.381	78.453	50 m	20 m		
	Mugguraikona	14.343	78.327	100 m	50 m		
Yellow ochre	Alavalapadu	14.420	78.386	40 m	10 m	a. Steep slope high wall b. Dangerous impoundment c. Loose surface	Effect on groundwater quality and soil quality
	Nagur	14.447	78.408	20 m	15 m		
	Ammayagari palli	14.408	78.443	20 m	25 m		
	Chagaleru	14.397	78.363	10 m	5 m		
	Ramanuthala palli	14.465	78.136	35 m	10 m		
Asbestos	Brahmanapalli	14.416	78.185	200 m	10 m	a. Steep high walls b. Steep portal mine openings c. Dangerous processing equipment facilities d. Air pollution	Effect on groundwater quality and soil quality, Acid mine drainage
White clay	Lingala	14.482	78.115	300 m	10 m	a. Steep slope high wall b. Dangerous impoundment c. Loose surface	Effect on groundwater quality and soil quality

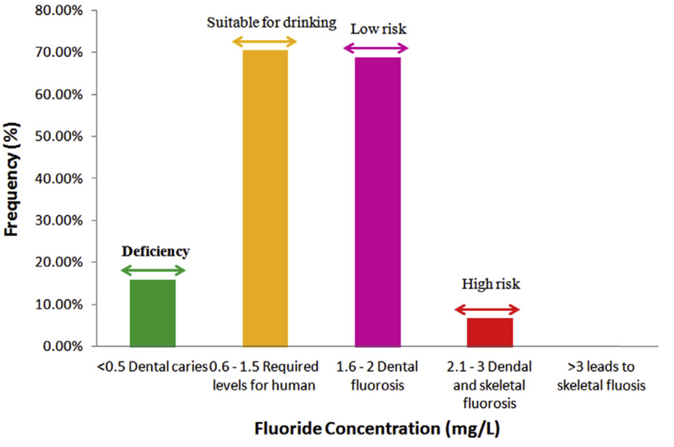

Fig. 4. Frequency distribution of fluoride concentration for risk evaluation in groundwater SW part of Cuddapah basin.

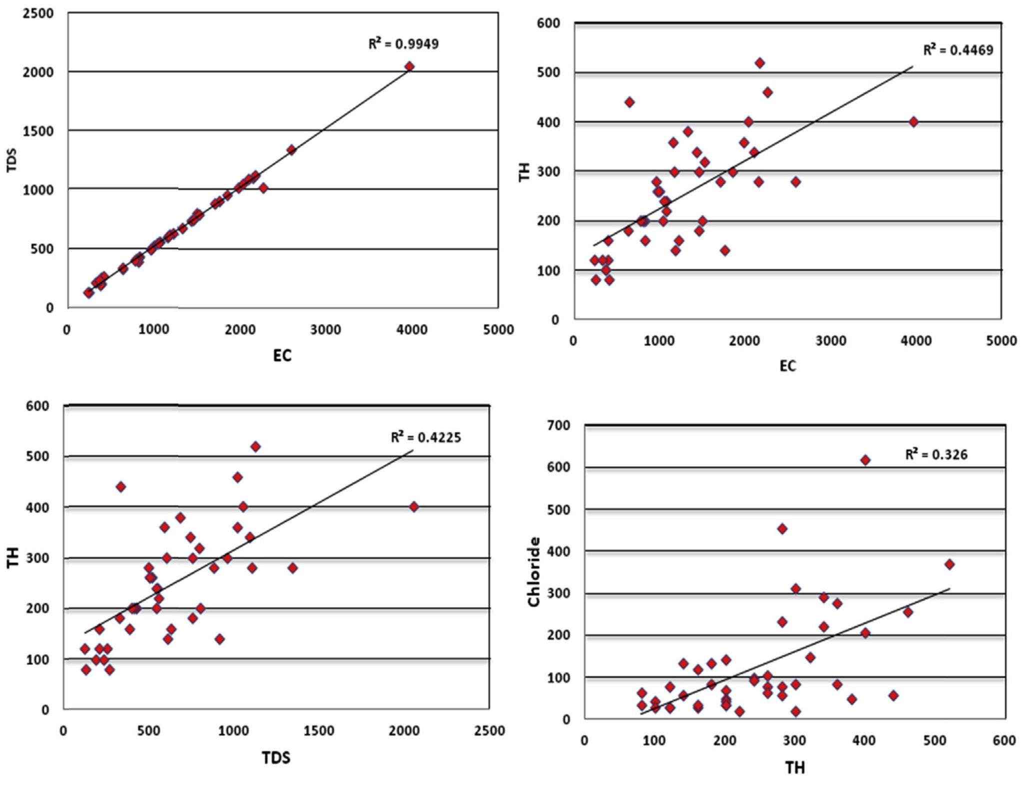

Fig. 5. Correlation among physico - chemical parameters of the Groundwater samples.

analyzer 371 field kit respectively; Total Hardness, Ca^{2+}, Mg^{2+}, CO_3^{2-}, HCO_3^- and Cl^- were determined using titrimetry as laboratory followed standard methods (APHA 1998); F^- is determined using ion-selective electrode (Orion 4 star ion meter, Model: pH/ISE).

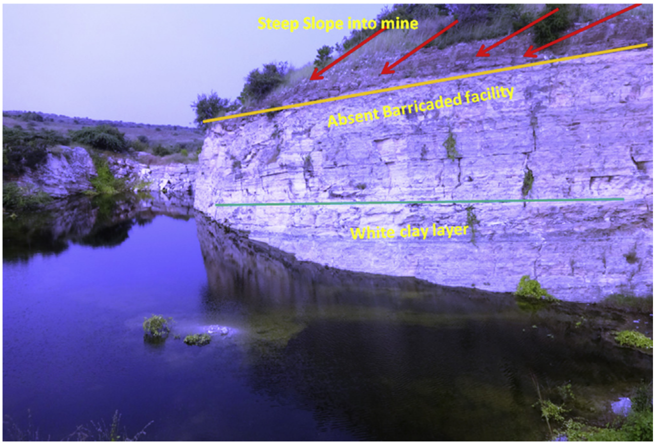

Fig. 6. Abandoned Clay mine near Lingala Village, Y·S.R district.

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Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.dib.2020.105187.

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